Yijie Pan

Ph.D. Candidate in Mathematics, University of Southern California

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RESEARCH INTERESTS

Symplectic and contact topology, Reeb dynamics, symplectic capacities, and gauge theory.

EDUCATION

2024–present **University of Southern California**, Ph.D. in Mathematics
Advisor: Julian Chaidez

Dec. 2023 **University of California, Los Angeles**, B.S. in Mathematics

PAPERS AND PREPRINTS

1. *Pseudo-Anosov Reeb flows and contact structures*, with Julian Chaidez. Preprint, 2025. arXiv:2512.12698.
2. *Polytopal counterexamples to the equality case of Viterbo's conjecture*, with Osagie Osayande and Henry Riley. Preprint, 2026.

WORK IN PROGRESS

1. *On the gluing formula of the Bauer–Furuta invariant*, with Baran Çetin.
2. *Symplectic TQFT for stable Hamiltonian structures*, with Michael Huang.

TALKS

Oct. 2026 *Pseudo-Anosov Reeb flows and contact structures*
AMS Fall Eastern Sectional Meeting, Special Session on Recent Developments at the Interface of Hyperbolic Dynamics and Geometric Structures, George Washington University, Washington, DC

Sep. 2026 *Crash course in analysis: pseudo-holomorphic curves and somewhere injectivity*
USC Topology Seminar on Symplectic Field Theory

Nov. 2025 *The classical Bauer–Furuta invariant*
Gauge Theory Seminar, University of Southern California

Jan. 2024 *Contact manifolds, I*
USC Student Topology Seminar

SERVICE

Fall 2026 Co-organizer, USC Topology Seminar on Symplectic Field Theory (with Louis-Simon Cyr and Michael Huang)

EXPOSITORY NOTES

Sep. 2026	*Pseudo-holomorphic curves and somewhere injectivity* — talk notes, USC Topology Seminar on Symplectic Field Theory
Nov. 2025	*The classical Bauer–Furuta invariant* — talk notes, USC Gauge Theory Seminar
Dec. 2024	*Representations of $\mathfrak{sp}_4(\mathbb{C})$*
Nov. 2023	*Atiyah–Hirzebruch's counterexample to the integral Hodge conjecture*
Nov. 2023	*K-theory and the Brouwer fixed point theorem*

Last updated October 2026